C#_Day2_Hands_On_Koduru_Ushasree

Level-1 Problem 1: Simple Calculator Using Methods

Scenario:

A small retail shop wants a simple calculator application to perform addition and subtraction operations using reusable methods.

Requirements:

1. Create a class named Calculator.
2. Create methods Add(int a, int b) and Subtract(int a, int b).
3. Each method should return the result.
4. In Main(), create an object and call the methods.
5. Display the output.

Technical Constraints:

1. Use method parameters and return types properly.
2. Use appropriate access modifiers.
3. No global variables allowed.

Expectations:

Proper method definition, object creation, and method invocation.

Learning Outcome:

Understand functions, parameters, return types, classes, and objects.

Sample Input: 10 5

Sample Output: Addition = 15, Subtraction = 5

Enhance the problem 1 by adding private data members, set values those data members and display data using public without using get set properties and without constructors.

Code :-

```
using System;
using System.Reflection.Emit;

namespace Day2_Problem1
{
    2 references
    public class Calculator
    {
        private int n1;
        private int n2;

        1 reference
        public int Add(int n1, int n2)
        {
            this.n1 = n1;
            this.n2 = n2;
            return this.n1 + this.n2;
        }
        1 reference
        public int Subtract(int n1, int n2)
        {
            this.n1 = n1;
            this.n2 = n2;
            return this.n1 - this.n2;
        }
    }
    0 references
```

```
    class problem2
    {
        0 references
        static void Main(string[] args)
        {
            Console.WriteLine("Enter first number:");
            int a = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("Enter second number:");
            int b = Convert.ToInt32(Console.ReadLine());

            Calculator calc = new Calculator();


            Console.WriteLine("Addition = " + calc.Add(a, b));
            Console.WriteLine("Subtraction = " + calc.Subtract(a, b));
        }
    }
}
```

Output :-

```
Microsoft Visual Studio Debu
Enter first number:
40
Enter second number:
20
Addition = 60
Subtraction = 20
```

Technical Requirements :-

The program should implement a **Calculator class** that contains private data members to store two numbers. It should provide **public methods for addition and subtraction**, where the values are passed as parameters and assigned to the class variables using the this keyword. The methods should perform the required calculation and return the result. In the Main() method, the program should **accept user input, create an object of the Calculator class, call the methods**, and display the calculated results on the console.